

Supplementary Material: Recording-Grouped Bearing Fault Diagnosis under Directional Operating-Condition Shifts on the MCC5 Electric-Drive Benchmark

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1 Preprocessing and Split Audits

All final neural and fusion runs use frozen source-recording roles. No source recording or window crosses training, validation, and test roles, and no window-level validation fallback is used. Standardization statistics are fitted on training data only. The audit contains 1425 checks and all pass.

Table S1: Normalization audit by publication-facing input definition and code identifier.

Publication label	Code identifier	Audited	Pass	Fail
Strict-254 scalar-branch fusion	full_engineered_features_254	1040	1040	0
Legacy-262 complete branch	full_multisource_fusion	268	268	0
Raw CNN/TCN/Transformer, 8,192 samples	raw_8192	24	24	0
Raw CNN length sensitivity, 12,800 samples	raw_length_12800	6	6	0
Raw CNN length sensitivity, 8,192 samples	raw_length_8192	6	6	0
Vibration-current; no scalars	vibration_current	7	7	0
Vibration-current + auxiliary-26	vibration_current_auxiliary_only	32	32	0
Vibration-current + auxiliary-context-28	vibration_current_rpm_load	34	34	0
Vibration-current + RPM/load	vibration_current_rpm_load_only	8	8	0

2 Complete Fusion-Input Evidence

The 26- and 28-field auxiliary definitions contain two modality-presence indicators and 24 torque/key-phase waveform summaries, with the 28-field configuration additionally including RPM and load. These two definitions were isolated after the 3000-rpm holdout had been inspected and are therefore post-hoc exploratory throughout this supplement. Their numerical detail is disclosed for transparency, not as confirmatory model selection evidence. The strict 254-feature and legacy 262-field branches are retained to document input-definition sensitivity.

Table S2: Complete fusion-input results across the four recording-grouped protocols. Values for source-recording, cross-mode, and cross-load are window-level macro-F1 means and standard deviations across three optimization seeds. The focused 1000/2000-to-3000-rpm column uses five optimization seeds. Auxiliary-26 and auxiliary-context-28 were identified after inspection of the same 3000-rpm holdout and are therefore post-hoc exploratory rather than confirmatory comparisons.

Configuration	Scalars	Source-recording	Cross-mode	Cross-load	1000/2000 to 3000 rpm
<i>Predefined matched configurations</i>					
Vibration-current; no scalars	0	0.8991 $\pm$ 0.0029	0.9417 $\pm$ 0.0062	0.8286 $\pm$ 0.0477	0.7081 $\pm$ 0.0986
Vibration-current + RPM/load	2	0.8455 $\pm$ 0.0375	0.9542 $\pm$ 0.0144	0.8309 $\pm$ 0.0885	0.7124 $\pm$ 0.0529
Strict-254 scalar-branch fusion	254	0.9939 $\pm$ 0.0011	0.9723 $\pm$ 0.0046	0.9299 $\pm$ 0.0350	0.4001 $\pm$ 0.0813
Legacy-262 complete branch	262	0.9943 $\pm$ 0.0012	0.9660 $\pm$ 0.0023	0.9571 $\pm$ 0.0266	0.4557 $\pm$ 0.0784
<i>Post-hoc exploratory configurations on the same holdout</i>					
Vibration-current + auxiliary-26*	26	0.9940 $\pm$ 0.0013	0.9820 $\pm$ 0.0086	0.9109 $\pm$ 0.0294	0.9162 $\pm$ 0.0245
Vibration-current + auxiliary-context-28*	28	0.9919 $\pm$ 0.0016	0.9884 $\pm$ 0.0154	0.9275 $\pm$ 0.0349	0.9095 $\pm$ 0.0284

* Post-hoc configurations identified after inspection of the same 3000-rpm holdout.

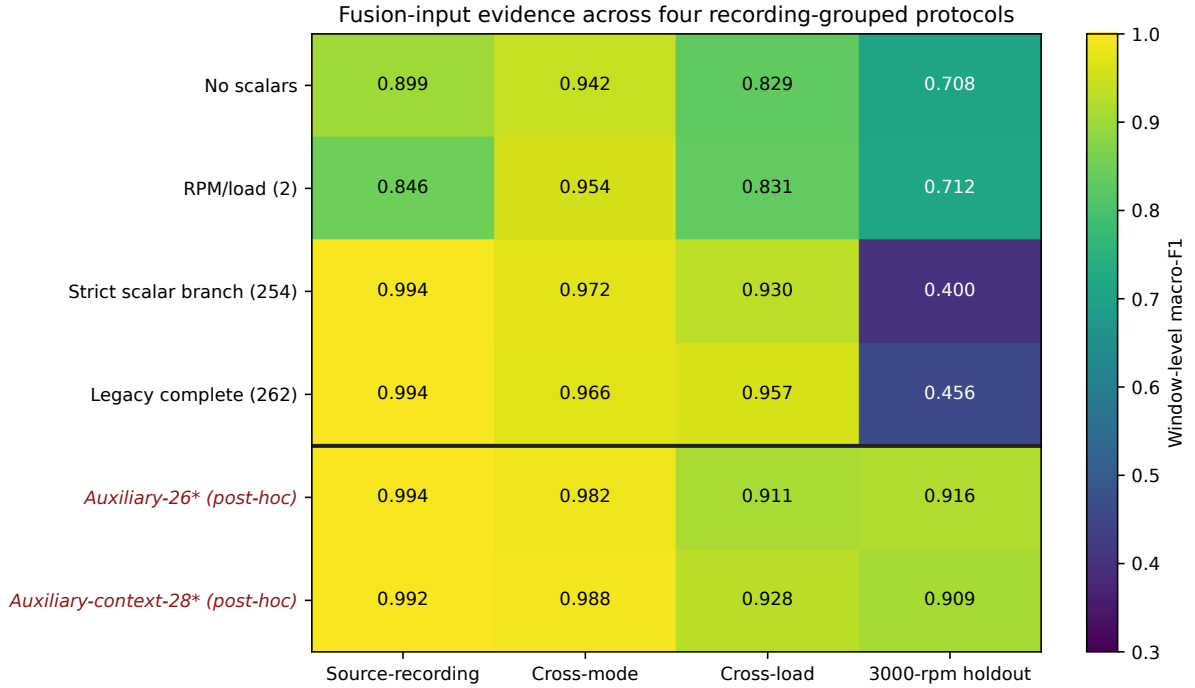


Figure S1: Complete four-protocol fusion-input comparison. The horizontal separator distinguishes predefined matched configurations from auxiliary-26* and auxiliary-context-28*, which are post-hoc exploratory configurations selected after inspection of the same 3000-rpm holdout. The displayed color range is 0.30–1.00.

2.1 Focused 3000-rpm evidence

Table S3: Per-seed fusion-input results for the 3000-rpm holdout. The auxiliary-26 and auxiliary-context-28 rows are post-hoc exploratory on this same holdout.

Configuration	Scalars	Seed	Macro-F1	Worst recall
<i>Predefined matched configurations</i>				
Vibration-current; no scalars	0	42	0.5599	0.2451
Vibration-current; no scalars	0	43	0.8027	0.6536
Vibration-current; no scalars	0	44	0.7366	0.4085
Vibration-current; no scalars	0	45	0.7792	0.3715
Vibration-current; no scalars	0	46	0.6623	0.4832
Vibration-current + RPM/load	2	42	0.7368	0.4714
Vibration-current + RPM/load	2	43	0.6231	0.2793
Vibration-current + RPM/load	2	44	0.7605	0.4874
Vibration-current + RPM/load	2	45	0.7116	0.3827
Vibration-current + RPM/load	2	46	0.7299	0.5154
Strict-254 scalar-branch fusion	254	42	0.5236	0.0223
Strict-254 scalar-branch fusion	254	43	0.3906	0.0279
Strict-254 scalar-branch fusion	254	44	0.4279	0.0321
Strict-254 scalar-branch fusion	254	45	0.3166	0.0251
Strict-254 scalar-branch fusion	254	46	0.3419	0.0209
Legacy-262 complete branch	262	42	0.4965	0.0251
Legacy-262 complete branch	262	43	0.5311	0.1648
Legacy-262 complete branch	262	44	0.4592	0.1243
Legacy-262 complete branch	262	45	0.3249	0.0223
Legacy-262 complete branch	262	46	0.4666	0.0880
<i>Post-hoc exploratory configurations on the same holdout</i>				
Vibration-current + auxiliary-26*	26	42	0.9121	0.8059
Vibration-current + auxiliary-26*	26	43	0.9431	0.8765
Vibration-current + auxiliary-26*	26	44	0.9274	0.8715
Vibration-current + auxiliary-26*	26	45	0.9209	0.8038
Vibration-current + auxiliary-26*	26	46	0.8773	0.7318
Vibration-current + auxiliary-context-28*	28	42	0.8819	0.6969
Vibration-current + auxiliary-context-28*	28	43	0.8820	0.6997
Vibration-current + auxiliary-context-28*	28	44	0.9429	0.8839
Vibration-current + auxiliary-context-28*	28	45	0.9336	0.8080
Vibration-current + auxiliary-context-28*	28	46	0.9070	0.8233

* Post-hoc configurations identified after inspection of the same 3000-rpm holdout.

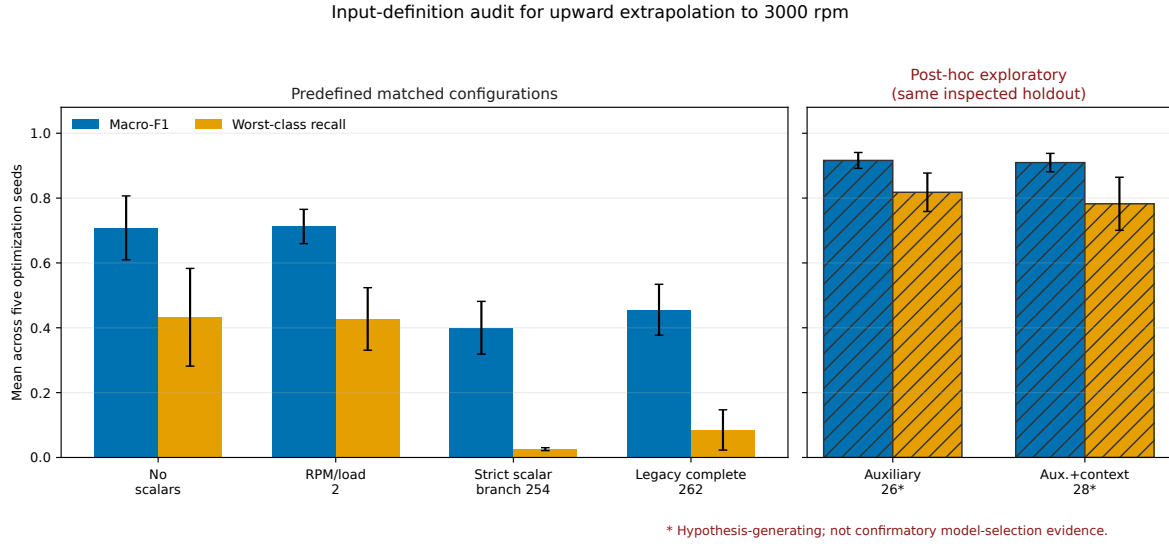


Figure S2: Focused fusion-input audit for upward extrapolation to 3000 rpm. The left panel contains predefined matched configurations; the separate hatched right panel contains post-hoc exploratory configurations identified on the same holdout. Error bars denote standard deviations across five optimization seeds. The right-panel comparison is hypothesis-generating rather than confirmatory.

Table S4: Recording-level mean-probability aggregation for all fusion-input definitions on the 3000-rpm holdout. Means and standard deviations are calculated across optimization seeds after probabilities have first been aggregated within each recording.

Configuration	Recordings	Recording macro-F1 mean $\pm$ SD	Recording worst recall mean $\pm$ SD	Evidence status
Vibration-current; no scalars	28	0.7122 $\pm$ 0.1238	0.4000 $\pm$ 0.2054	Matched protocol
Vibration-current + RPM/load	28	0.7306 $\pm$ 0.0917	0.4000 $\pm$ 0.1369	Matched protocol
Strict-254 scalar-branch fusion	28	0.3660 $\pm$ 0.1254	0.0000 $\pm$ 0.0000	Matched protocol
Legacy-262 complete branch	28	0.3829 $\pm$ 0.0917	0.0000 $\pm$ 0.0000	Matched protocol
Vibration-current + auxiliary-26*	28	0.9435 $\pm$ 0.0264	0.8361 $\pm$ 0.0669	Post-hoc exploratory
Vibration-current + auxiliary-context-28*	28	0.9368 $\pm$ 0.0513	0.8278 $\pm$ 0.1439	Post-hoc exploratory

* Post-hoc exploratory configuration identified after inspection of the same 3000-rpm holdout.

3 Severity and Per-Recording Probability Audits

The severity analysis is a descriptive, low-cost audit derived from saved source-recording predictions. It was not used for model selection. High- and light-severity fault recordings are evaluated separately; healthy recordings remain ungraded.

Table S5: Severity-stratified recording-level results on the 3000-rpm holdout. Severity is inferred from the public recording identifier; healthy recordings have no severity grade. Values are means and standard deviations across optimization seeds.

Configuration	Severity	Recordings	Accuracy mean $\pm$ SD	Fault macro-F1 mean $\pm$ SD	Evidence status
Vibration-current; no scalars	High	12	0.6333 ± 0.1728	0.5771 ± 0.2166	Matched protocol
Vibration-current; no scalars	Light	12	0.8667 ± 0.1264	0.8631 ± 0.1303	Matched protocol
Vibration-current; no scalars	Healthy (ungraded)	4	0.6000 ± 0.3354	Not applicable (single class)	Matched protocol
Vibration-current + RPM/load	High	12	0.6667 ± 0.1768	0.6272 ± 0.2172	Matched protocol
Vibration-current + RPM/load	Light	12	0.9333 ± 0.0373	0.9323 ± 0.0379	Matched protocol
Vibration-current + RPM/load	Healthy (ungraded)	4	0.5000 ± 0.2500	Not applicable (single class)	Matched protocol
Strict-254 scalar-branch fusion	High	12	0.4000 ± 0.1369	0.3600 ± 0.1710	Matched protocol
Strict-254 scalar-branch fusion	Light	12	0.7500 ± 0.1559	0.6929 ± 0.1965	Matched protocol
Strict-254 scalar-branch fusion	Healthy (ungraded)	4	0.0000 ± 0.0000	Not applicable (single class)	Matched protocol
Vibration-current + auxiliary-26*	High	12	0.8500 ± 0.0697	0.8474 ± 0.0725	Post-hoc exploratory
Vibration-current + auxiliary-26*	Light	12	1.0000 ± 0.0000	1.0000 ± 0.0000	Post-hoc exploratory
Vibration-current + auxiliary-26*	Healthy (ungraded)	4	1.0000 ± 0.0000	Not applicable (single class)	Post-hoc exploratory

Fault macro-F1 is calculated over the three fault-location classes within each severity subgroup; healthy recordings are reported separately and therefore have no fault macro-F1. * Post-hoc exploratory configuration identified after inspection of the same 3000-rpm holdout.

Because severity is inferred from recording identifiers and the subgroups are small, these results are descriptive only. The observed differences should not be interpreted as evidence that light-severity faults are intrinsically easier to diagnose than high-severity faults.

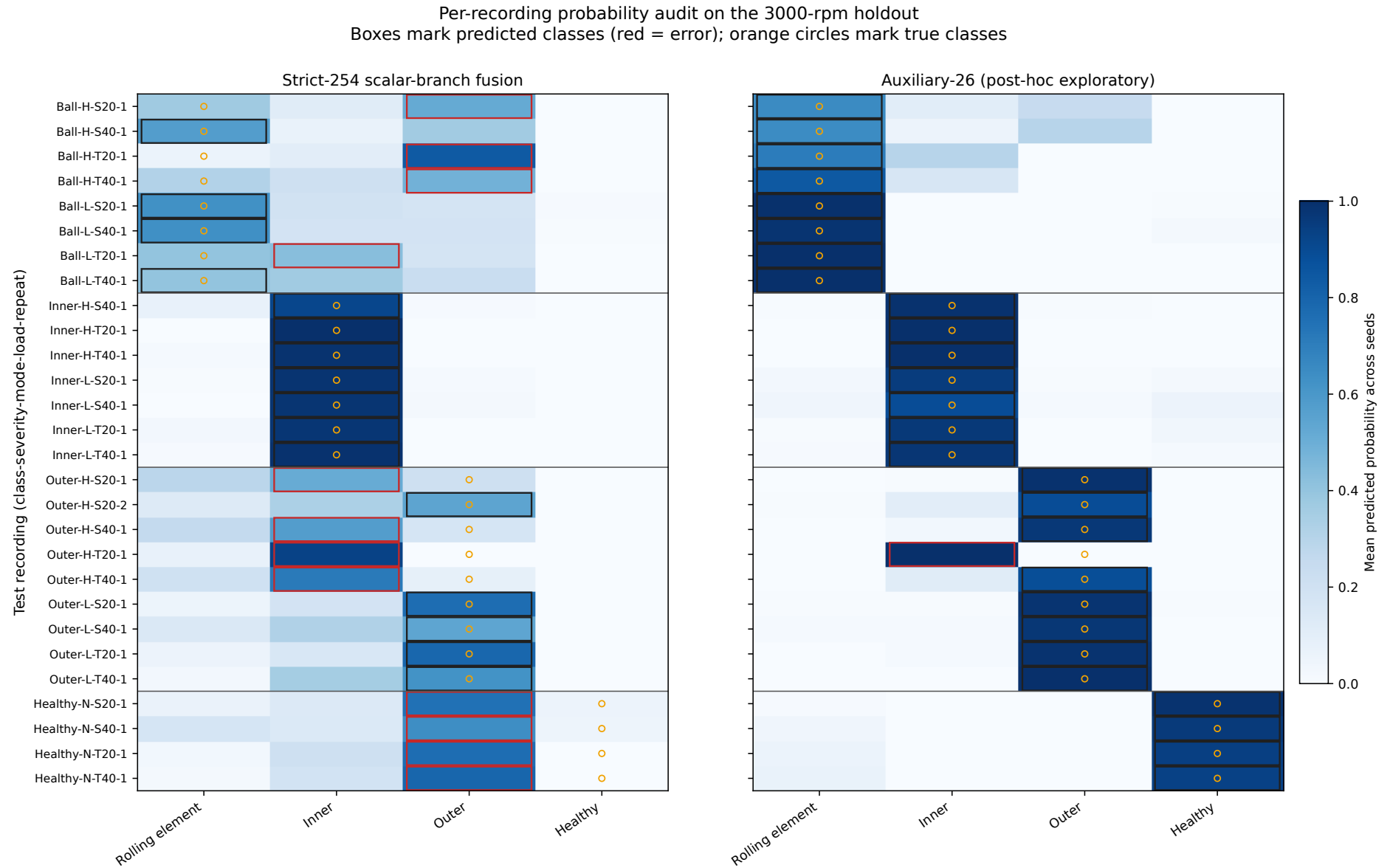


Figure S3: Per-recording class-probability audit on the 3000-rpm holdout, averaged across optimization seeds. Boxes mark the argmax class of the seed-averaged probability vector; red boxes denote errors, and orange circles denote the true class. Publication-facing class labels use “rolling element”; original recording identifiers retain the dataset prefix “Ball”. The auxiliary-26 panel is post-hoc exploratory on the same holdout and is shown only to localize the associated recording-level behavior.

4 Recording-Level Bootstrap

Intervals resample complete test recordings within class for 2,000 replicates. They quantify uncertainty over the finite set of test recordings and do not convert post-hoc model selection into confirmatory evidence.

Table S6: Class-stratified recording bootstrap for selected 3000-rpm models. Point estimates are recording-level macro-F1 values obtained after mean-probability aggregation within each recording. Intervals resample complete recordings within class and condition on the listed trained seed. The Auxiliary-26 row is post-hoc exploratory, and the intervals do not remove its selection bias.

Model	Evidence status	Seed	Point macro-F1	95% interval	Recordings
Strict-254 scalar-branch fusion	Matched protocol	42	0.5348	[0.4242, 0.6244]	28
Strict-254 scalar-branch fusion	Matched protocol	43	0.2906	[0.2101, 0.3713]	28
Strict-254 scalar-branch fusion	Matched protocol	44	0.4478	[0.3439, 0.5519]	28
Strict-254 scalar-branch fusion	Matched protocol	45	0.2208	[0.1540, 0.2955]	28
Strict-254 scalar-branch fusion	Matched protocol	46	0.3359	[0.2189, 0.4581]	28
Auxiliary-26 branch*	Post-hoc exploratory	42	0.9374	[0.8453, 1.0000]	28
Auxiliary-26 branch*	Post-hoc exploratory	43	0.9686	[0.9059, 1.0000]	28
Auxiliary-26 branch*	Post-hoc exploratory	44	0.9686	[0.9059, 1.0000]	28
Auxiliary-26 branch*	Post-hoc exploratory	45	0.9375	[0.8381, 1.0000]	28
Auxiliary-26 branch*	Post-hoc exploratory	46	0.9055	[0.7967, 1.0000]	28
Raw vibration-current CNN	Matched protocol	42	0.4367	[0.2922, 0.5515]	28
Raw vibration-current CNN	Matched protocol	43	0.8888	[0.7297, 1.0000]	28
Raw vibration-current CNN	Matched protocol	44	0.8444	[0.7134, 0.9375]	28

5 Model Architectures and Fixed Hyperparameters

Table S7 reports the publication-facing settings implemented in the formal benchmark. Classical hyperparameters were fixed before final test evaluation. Neural early stopping and checkpoint selection used only the frozen internal-validation recordings; held-out test recordings did not inform scaling, checkpoint selection, or hyperparameter choice.

Table S7: Model architectures, preprocessing, and fixed hyperparameters.

Model	Input and preprocessing	Architecture and fixed hyperparameters	Selection and size
Logistic regression	254 engineered fields; training-median imputation and StandardScaler	C=1.0; L2 penalty; lbfgs solver; max_iter=1000	Fixed before test; 1,020 coefficients/intercepts for 254 inputs and four classes
RBF-SVM	254 engineered fields; training-median imputation and StandardScaler	RBF kernel; C=3.0; gamma=scale	Fixed before test; support-vector count is training-data dependent
Random forest	254 engineered fields; training-median imputation; no scaling	200 trees; max_depth=None; min_samples_leaf=1; all CPU cores	Fixed before test; trained tree size is data dependent
MLP	254 engineered fields; training-median imputation and StandardScaler	254->128->4; ReLU; Adam; alpha=1e-4; max_iter=300	Fixed before test; 33,156 trainable weights and biases

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Table S7 continued

Model	Input and preprocessing	Architecture and fixed hyperparameters	Selection and size
XGBoost	254 engineered fields; training-median imputation; no scaling	200 trees; max_depth=4; learning_rate=0.05; subsample=0.9; colsample_bytree=0.9; multiclass log-loss	Fixed before test; trained tree size is data dependent
Raw CNN	Six vibration-current channels; 8,192 samples; train-only per-channel z-score	Conv1d 6->32 (kernel 9), batch normalization, ReLU, max-pool 4; Conv1d 32->64 (kernel 7), batch normalization, ReLU, global average pooling; linear 64->4	Best validation macro-F1; 16,612 trainable parameters
TCN	Six vibration-current channels; 8,192 samples; train-only per-channel z-score	Three non-residual dilated Conv1d blocks, width 32, kernel 5, dilations 1/2/4; batch normalization and ReLU; global average pooling; linear 32->4	Best validation macro-F1; 11,620 trainable parameters
Transformer	Six vibration-current channels; 8,192 samples; train-only per-channel z-score	Conv1d token projection 6->64 (kernel 9, stride 8); two encoder layers; d_model=64; four heads; feed-forward width 128; dropout=0.1; mean pooling; linear 64->4	Best validation macro-F1; 70,724 trainable parameters
Fusion family	Separate three-channel vibration and current sequences plus 0/2/26/28/254/262 scalar definitions; train-only sequence and scalar z-score	Each sequence encoder: Conv1d 3->32 (kernel 9), max-pool 4, Conv1d 32->64 (kernel 7), global average pooling. Scalar MLP d->32->32. Concatenated width 160, sigmoid gate 160->160, classifier 160->64->4, dropout=0.2	Best validation macro-F1; parameters: 68,420 / 68,452 / 69,220 / 69,284 / 76,516 / 76,772 in the listed scalar order

Classical configurations were fixed before final test evaluation. Neural checkpoints were selected by the highest internal-validation macro-F1. Neural models used class-weighted cross-entropy, Adam with learning rate 10^{-3} , batch size 64, at most 30 epochs, and patience 8; fusion models additionally used weight decay 10^{-4} .

6 Reproducibility Notes

The verified runtime used Python 3.12.13, PyTorch 2.11.0+cu128, and an NVIDIA GeForce RTX 5080. Automated tests cover train-only sequence and scalar standardization, constant channels, missing scalar values, serialization of fitted statistics, source-recording separation across all data roles, and the evidence-status labels applied to exploratory configurations. The software supplement includes frozen split maps, a de-identified window-to-recording index, exact input schemas, result tables, and reproduction commands; raw MCC5 signals are not redistributed.